2026 `src/scx/`
Target: Synthese / Philosophy of Science / Erkenntnis

SCX

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State-Conditioned eXpertise and Scientific Consensus:
A Formalization of Philosophy of Science

SCX

July 2, 2026

Abstract

—— —— SCXState-Conditioned eXpertise $M\mathcal{S}$
M Chernoff-Hoeffding$\Delta_s > 0M \geq M_{\text{crit}}$ s $M\exp(-2M\Delta_s^2)$
Kuhn$\mathcal{S}_K$ $\mathcal{E}_K D_{\text{new}}$Fano—— $I(Y; X_{\text{new}} \mid \mathcal{S}_K) > H(Y \mid \mathcal{S}_K) - \log 2$——
SCXPopperSCX s $(H_0 : \Delta_s \leq 0,\ H_1 : \Delta_s > 0)$ $\Delta_s^{\text{claimed}}\mathcal{V}$ $\Delta_s^{\text{claimed}}$
SCX $\{E'_1, \ldots, E'_{M'}\}$ $\hat{\Delta}_s^{\text{rep}} \leq 0$Chernoff SCX
Honest Critique//
SCXChernoff-HoeffdingFano

1

1.1

20Popper KuhnLakatosFeyerabend

——

(1) Cover & Thomas (2006)Wainwright (2019) Boucheron(2013)

(2) SCXYajie(SCX Project, 2026) “”

(3) Open Science Collaboration (2015) Ioannidis (2005)“”

SCX————
M Δ_s Chernoff-HoeffdingFano

1.2 SCX

SCX

Definition 1.1 (——)**.** $\mathcal{S} = \{s_1, s_2, \ldots, s_N\}$ s_i

(i) $\phi(s_i) \in \mathbb{R}^d$

(ii) $y_i \in \mathcal{Y}//$

(iii) $/p_i \in \mathcal{P}$

Definition 1.2 (——). $M\mathcal{E} = \{E_1, E_2, \ldots, E_M\}$ sm

$$v_m(s) = \mathbf{1}[E_m(\phi(s)) \neq y] \in \{0, 1\} \tag{1}$$

$\mathbf{1}[\cdot]$
$v_m(s) = 0 s v_m(s) = 1$
“”

Axiom 1.3 (——). y_s

$$v_m(s) \perp\!\!\!\perp v_{m'}(s) \mid y_s, \quad \forall m \neq m' \tag{2}$$

[] —— §6.1

Axiom 1.4 (——). $y_s v_m(s) = 1 p_{clean,s} < 0.5$ $y_s v_m(s) = 1 p_{noisy,s} > 0.5$

$$\boxed{\Delta_s = p_{noisy,s} - p_{clean,s}} \tag{3}$$

$\Delta_s > 0$

Axiom 1.5 (——Fano). *Fano*

$$P_e \geq \frac{H(Y \mid X) - 1}{\log |\mathcal{Y}|} \tag{4}$$

$P_e H(Y \mid X) X Y$
——

2 M

2.1

——

Definition 2.1. $s \in \mathcal{S} M \{v_1(s), \ldots, v_M(s)\}$

$$C_M(s) = \frac{1}{M} \sum_{m=1}^{M} (1 - v_m(s)) = 1 - \frac{1}{M} \sum_{m=1}^{M} v_m(s) \tag{5}$$

s
$s\alpha$-$\tau_\alpha \in (0.5, 1]$

$$\boxed{\mathbb{P}\left(C_M(s) \geq \tau_\alpha \mid \Delta_s > 0\right) \geq 1 - \alpha} \tag{6}$$

$M 1 - \alpha$

$M \alpha \tau_\alpha$

2.2 ——

Theorem 2.2 (Honest Person Theorem). M**??**$\Delta_s > 0$ $M \to \infty C_M(s) 1 1 - p_{clean,s}$

$$\boxed{\mathbb{P}\left(\lim_{M \to \infty} C_M(s) = 1 - p_{clean,s}\right) = 1} \tag{7}$$

$\varepsilon > 0 C_M(s) \varepsilon$ *Chernoff-Hoeffding*

$$\boxed{\mathbb{P}\left(|C_M(s) - (1 - p_{clean,s})| \geq \varepsilon\right) \leq 2 \exp\left(-2M\varepsilon^2\right)} \tag{8}$$

Proof. **1**$y_s v_m(s) \sim \text{Bernoulli}(p_{\text{clean},s})$ $\frac{1}{M}\sum v_m(s) \xrightarrow{\text{a.s.}} p_{\text{clean},s}$ $C_M(s) = 1 - \frac{1}{M}\sum v_m(s) \xrightarrow{\text{a.s.}} 1 - p_{\text{clean},s}$

2Chernoff-Hoeffding$C_M(s)M[0,1]$ Hoeffding$\varepsilon > 0$

$$\mathbb{P}\left(|C_M(s) - \mathbb{E}[C_M(s)]| \geq \varepsilon\right) \leq 2\exp\left(-2M\varepsilon^2\right) \tag{9}$$

$\mathbb{E}[C_M(s)] = 1 - p_{\text{clean},s}$

32

$$\tau_\alpha = 1 - p_{\text{clean},s} - \sqrt{\frac{\log(2/\alpha)}{2M}} \tag{10}$$

$\mathbb{P}(C_M(s) \geq \tau_\alpha) \geq 1 - \alpha\alpha$- □ □

[] 12Hoeffding(1963) 3**????**
“” —— Condorcet

2.3 ——

“”Galileo, Wegener, Boltzmann

Theorem 2.3 (Unlikely-to-be-Solo-Awake Theorem)**.** *$MM-1s1$ $\Delta_s > 0$*

$$\boxed{\mathbb{P}\left(\mid C_M(s) = \frac{M-1}{M}, \Delta_s > 0\right) \leq \frac{p_{noisy,s}}{p_{noisy,s} + (M-1)(1-p_{clean,s}) \cdot \frac{p_{noisy,s}^{M-2}}{p_{clean,s}^{M-2}} \cdot \frac{1-p_{noisy,s}}{1-p_{clean,s}}}} \tag{11}$$

$M \to \infty$$\exp(-2M\Delta_s^2)$

Proof. **1Bayes**$H_0 s H_1 s$ $M-1 1$

$$\mathbb{P}(H_1 \mid \text{data}) = \frac{\mathbb{P}(\text{data} \mid H_1) \cdot \mathbb{P}(H_1)}{\mathbb{P}(\text{data} \mid H_0) \cdot \mathbb{P}(H_0) + \mathbb{P}(\text{data} \mid H_1) \cdot \mathbb{P}(H_1)} \tag{12}$$

2H_0 $\mathbb{P}(\text{data} \mid H_0) = p_{\text{clean},s}^{M-1} \cdot (1 - p_{\text{clean},s})M - 1 1$ H_1 $\mathbb{P}(\text{data} \mid H_1) = p_{\text{noisy},s} \cdot (1 - p_{\text{noisy},s})^{M-1} 1 M - 1$

3odds

$$\frac{\mathbb{P}(H_1 \mid \text{data})}{\mathbb{P}(H_0 \mid \text{data})} = \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)} \cdot \frac{p_{\text{noisy},s}}{1 - p_{\text{clean},s}} \cdot \left(\frac{1 - p_{\text{noisy},s}}{p_{\text{clean},s}}\right)^{M-1} \tag{13}$$

$p_{\text{clean},s} < 0.5 < p_{\text{noisy},s}\Delta_s > 0$ $1 - p_{\text{noisy},s} < 0.5 < p_{\text{clean},s}$ $(1 - p_{\text{noisy},s})/p_{\text{clean},s} < 1M$

4Chernoff Hoeffding

$$\mathbb{P}(H_1 \mid \text{data}) \leq \exp(-2M\Delta_s^2 + O(1)) \tag{14}$$

$\Delta_s = p_{\text{noisy},s} - p_{\text{clean},s} > 0$ □ □

[] 1–23 $\Delta_s > 0$4ChernoffHoeffding $\mathbb{P}(H_0) \approx \mathbb{P}(H_1)$ H_1**??**

Remark 2.4. (i) $\mathbb{P}(H_1)$1(ii) (i)p-hacking—— M(ii) $M_{\text{eff}} < M$§6.1

2.4 Chernoff

Theorem 2.5 (——Chernoff)**.** *$sM\alpha$ s*

$$\boxed{C_M(s) \geq \frac{1}{2} + \sqrt{\frac{\log(1/\alpha)}{2M}}} \tag{15}$$

$$\boxed{M \geq M_{crit} = \frac{\log(1/\alpha)}{2(C_M(s) - 1/2)^2}} \tag{16}$$

Proof. $H_0 : \Delta_s \leq 0$ $C_M(s) \leq 1/2$Hoeffding

$$\mathbb{P}_{H_0}\left(C_M(s) \geq \frac{1}{2} + t\right) \leq \exp(-2Mt^2) \tag{17}$$

$= \alpha t = \sqrt{\log(1/\alpha)/(2M)}$ $H_0 1/2 + \sqrt{\log(1/\alpha)/(2M)}$ $M M_{\text{crit}}$ □ □

[] **??** II$\Delta_s \beta \approx \exp(-2M(\Delta_s - t)^2)$ $\Delta_s 0 M$
Δ_s —— §3

3 KuhnFano

3.1 -

Thomas Kuhn (1962)“” “” Kuhn———— SCX Fano

Definition 3.1 (KuhnSCX)**.** ***Kuhn***$\mathcal{K}$

$$\boxed{\mathcal{K} = (\mathcal{S}_K, \mathcal{E}_K)} \tag{18}$$

- $\mathcal{S}_K \subset \mathcal{S}$—— “”
- $\mathcal{E}_K = \{E_1^K, \ldots, E_{M_K}^K\}$——

$$H_K(Y \mid \mathcal{S}_K) = H(Y) - I_K(Y; \mathcal{S}_K) \tag{19}$$

$I_K(Y; \mathcal{S}_K)\mathcal{K}$

Definition 3.2 (——)**.** $\mathcal{K} D_{new} = \{(x_i, y_i)\}_{i=1}^n$ ε-

$$\boxed{I(Y; D_{new} \mid \mathcal{S}_K) > \varepsilon} \tag{20}$$

ε

3.2 Fano

Theorem 3.3 (——Fano)**.** *$\mathcal{K} D_{new}$ Fano*

$$\boxed{I(Y; D_{new} \mid \mathcal{S}_K) > H(Y \mid \mathcal{S}_K) - \log 2} \tag{21}$$

$\mathcal{K}$ $\mathcal{E}_K 1/2$——

Proof. **1Fano**$\mathcal{K}$

$$P_e^{\mathcal{K}} \geq \frac{H(Y \mid \mathcal{S}_K) - 1}{\log |\mathcal{Y}|} \tag{22}$$

$/|\mathcal{Y}| = 2 \log |\mathcal{Y}| = \log 2$
2D_{new}

$$H(Y \mid \mathcal{S}_K, D_{\text{new}}) = H(Y \mid \mathcal{S}_K) - I(Y; D_{\text{new}} \mid \mathcal{S}_K) \tag{23}$$

3Fano

$$P_e^{\mathcal{K}+D_{\text{new}}} \geq \frac{H(Y \mid \mathcal{S}_K, D_{\text{new}}) - 1}{\log 2} = \frac{H(Y \mid \mathcal{S}_K) - I(Y; D_{\text{new}} \mid \mathcal{S}_K) - 1}{\log 2} \tag{24}$$

$\mathbf{4}P_e^{\mathcal{K}+D_{\text{new}}} > 1/2$

$$\frac{H(Y \mid \mathcal{S}_K) - I(Y; D_{\text{new}} \mid \mathcal{S}_K) - 1}{\log 2} > \frac{1}{2} \tag{25}$$

$$H(Y \mid \mathcal{S}_K) - I(Y; D_{\text{new}} \mid \mathcal{S}_K) - 1 > \frac{1}{2}\log 2 \tag{26}$$

$$H(Y \mid \mathcal{S}_K) - I(Y; D_{\text{new}} \mid \mathcal{S}_K) > 1 + \frac{1}{2}\log 2 \tag{27}$$

nat

$$I(Y; D_{\text{new}} \mid \mathcal{S}_K) > H(Y \mid \mathcal{S}_K) - \log 2 \tag{28}$$

□ □

[] Fano(Cover & Thomas, 2006, 2.10.1) 24
Kuhn“”
——

3.3

Theorem 3.4. $\mathcal{K}_{old}\mathcal{K}_{new}$

$$\boxed{G(\mathcal{K}_{new} \mid \mathcal{K}_{old}) = I(Y; \mathcal{S}_{new}) - I(Y; \mathcal{S}_{old})} \tag{29}$$

$r(t)$————

$$\boxed{r(t) \propto \frac{G(\mathcal{K}_{new} \mid \mathcal{K}_{old})}{H(\mathcal{S}_{new} \mid \mathcal{S}_{old})} \cdot \frac{1}{\tau_{gen}}} \tag{30}$$

τ_{gen}“” $H(\mathcal{S}_{new} \mid \mathcal{S}_{old})$

. **1**E_m $U(E_m; \mathcal{K})$ G→
2$H(\mathcal{S}_{\text{new}} \mid \mathcal{S}_{\text{old}})$ “” →
3Kuhn——“ ”——τ_{gen} /
[] $G/H1/\tau_{\text{gen}}$

□

Corollary 3.5 (Kuhn——Planck)**.** ***Planck*“ ”*SCX***

$$\boxed{t_{shift} = \tau_{gen} \cdot \frac{H(\mathcal{S}_{new} \mid \mathcal{S}_{old})}{G(\mathcal{K}_{new} \mid \mathcal{K}_{old})}} \tag{31}$$

t_{shift} $t_{shift} \gg \tau_{gen}$——

[] (i) $G(\mathcal{K}_{\text{new}})$ (ii) $H(\mathcal{S}_{\text{new}} \mid \mathcal{S}_{\text{old}})$ Kuhn

4 PopperSCX

4.1

Karl Popper (1935/1959) Popper

(1) **Duhem-Quine**——

(2) —— “> 70%”

SCX

4.2 SCX

Definition 4.1 (SCX——). *$s \in \mathcal{S}$**SCX-***

$$\boxed{\mathcal{F}(s) = (\Delta_s^{claimed}, \mathcal{V}, M_{min})} \tag{32}$$

- $\Delta_s^{claimed} > 0$—— M
- $\mathcal{V}$——
- M_{min}—— *Chernoff*$H_0 H_1$

Theorem 4.2 (——SCXPopper)**.** *sSCX-$\mathcal{V}M$ power $> 1/2$*

$$H_0 : \Delta_s \leq 0 \quad \text{——} \tag{33}$$
$$H_1 : \Delta_s \geq \Delta_s^{claimed} > 0 \quad \text{——} \tag{34}$$

*$\Delta_s^{claimed}\mathcal{V}$ s**SCX-**——*

Proof. **1**$\mathcal{F}(s)$

- $\Delta_s^{\text{claimed}} \to / \to$
- $\mathcal{V} \to \to$
- $M_{\text{min}} \to$ —— “”

2——Chernoff$\mathcal{F}(s)$ Chernoff

$$\text{Power}(M, \Delta_s^{\text{claimed}}, \alpha) = \mathbb{P}_{H_1}\left(C_M(s) \geq \frac{1}{2} + \sqrt{\frac{\log(1/\alpha)}{2M}}\right) \geq 1-\exp\left(-2M\left(\Delta_s^{\text{claimed}} - \sqrt{\frac{\log(1/\alpha)}{2M}}\right)^2\right) \tag{35}$$

$M \geq M_{\text{min}} M_{\text{min}} > 1/2$

3Duhem-QuineSCX$A_1, \ldots, A_k$SCX $\phi(s)\mathcal{V}$ H_0SCX“” ——Duhem-Quine

4SCX“$p > 0.7$” Δ_sBernoulli $\Delta_s = p - (1-p) = 2p - 1 p > 0.5$ □ □

[] Hoeffding Duhem-Quine3“” $I(Y; A_j \mid S, \{A_k\}_{k \neq j})$ []

4.3

Theorem 4.3 (SCX)**.** *SCX*

(UF1) *$\Delta_s^{claimed}$—— “”$\Delta_s \to$Chernoff*

(UF2) *$\mathcal{V}$—— “”$\to$*

(UF3) *M_{min}—— “”$\lim_{M\to\infty}$*

sSCX-

Proof. **??**(1)$\mathcal{F}(s)$ □ □

[] **SCX** SCX

(1) $\mathcal{V}$ ——

(2) $\Delta_s^{\textbf{claimed}}$ $\Delta_s^{\text{claimed}}$ M

(3) $\mathcal{S}$ ——Kuhn-FeyerabendSCX

SCX SCX

5 SCX

5.1 SCX

replication crisis Open Science Collaboration (2015)36% Camerer(2018)62%Errington(2021) 46% SCX

Definition 5.1 (——SCX). *$s\mathcal{E}_{orig} = \{E_1, \ldots, E_M\}$ $\mathcal{E}_{rep} = \{E'_1, \ldots, E'_{M'}\}$ $\mathcal{V}SCX$*

$$\boxed{\hat{\Delta}_s^{rep} > 0 \qquad \left|\hat{\Delta}_s^{rep} - \hat{\Delta}_s^{orig}\right| \leq \sqrt{\frac{\log(2/\alpha)}{2}\left(\frac{1}{M} + \frac{1}{M'}\right)}} \tag{36}$$

$\hat{\Delta}_s^{orig}\hat{\Delta}_s^{rep}$

5.2 SCX

Theorem 5.2. *SCX*

1. ***Margin Reversal*** $\hat{\Delta}_s^{rep} \leq 0$ $\hat{\Delta}_s^{orig} > 0$
2. ***Expert Collapse*** $\hat{\Delta}_s^{rep} \approx \hat{\Delta}_s^{orig} > 0$ $M'_{\mathit{eff}} \ll M'_{nominal}$——
3. ***State Occlusion*** $\hat{\Delta}_s^{rep}(\mathcal{S}') > 0$ $\hat{\Delta}_s^{rep}(\mathcal{S}) \leq 0$——

Proof. **1??** $\hat{\Delta}_s^{\mathrm{rep}} > 0C_{M'}(s)1/2$ Chernoff**??** Δ_sp-hacking

2$\mathcal{E}_{\mathrm{rep}}$ $M'_{\mathrm{eff}} < M'_{\mathrm{nominal}}$ Chernoff-HoeffdingJanson, 2004 $\hat{\Delta}_s^{\mathrm{rep}}O(1/M'_{\mathrm{eff}})O(1/M')$ $M'_{\mathrm{eff}} \ll M'_{\mathrm{nominal}}$ “” “”

3$\mathcal{S}$ $\mathcal{S}' \subset \mathcal{S}$ $\mathcal{S} \setminus \mathcal{S}'$→

$$\hat{\Delta}_s^{\mathrm{rep}}(\mathcal{S}) = \frac{|\mathcal{S}'|}{|\mathcal{S}|}\hat{\Delta}_s^{\mathrm{rep}}(\mathcal{S}') + \frac{|\mathcal{S} \setminus \mathcal{S}'|}{|\mathcal{S}|}\hat{\Delta}_s^{\mathrm{rep}}(\mathcal{S} \setminus \mathcal{S}') \tag{37}$$

$\hat{\Delta}_s^{\mathrm{rep}}(\mathcal{S} \setminus \mathcal{S}') \leq 0$ $|\mathcal{S}'|/|\mathcal{S}|$ “X”X □ □

[] $\hat{\Delta}_s$ Hoeffding

5.3

Theorem 5.3.

$$\boxed{I(Y; \mathcal{E}_{orig}) + I(Y; \mathcal{E}_{rep} \mid \mathcal{E}_{orig}) \leq H(Y)} \tag{38}$$

$I(Y; \mathcal{E}_{orig})$ $I(Y; \mathcal{E}_{rep} \mid \mathcal{E}_{orig})$ ——

Proof.

$$I(Y; \mathcal{E}_{\mathrm{orig}}, \mathcal{E}_{\mathrm{rep}}) = I(Y; \mathcal{E}_{\mathrm{orig}}) + I(Y; \mathcal{E}_{\mathrm{rep}} \mid \mathcal{E}_{\mathrm{orig}}) \leq H(Y) \tag{39}$$

$I(Y; \mathcal{E}_{\mathrm{orig}}) \approx H(Y)$ “” $I(Y; \mathcal{E}_{\mathrm{rep}} \mid \mathcal{E}_{\mathrm{orig}}) \approx 0$ —— □ □

[] **??** ——
M “”—— “”NSCX

Corollary 5.4 (SCX).

$$\boxed{I(Y; \mathcal{E}_{rep} \mid \mathcal{E}_{orig})} \tag{40}$$

——

6

6.1

??

(1) $\rightarrow$

(2) **Monoculture** R$\rightarrow$“”

(3) $\rightarrow$“”

(4) Kuhn$\rightarrow$ Δ_s

Theorem 6.1 (——)**.** *Chernoff-HoeffdingM* M_{eff}

$$\boxed{M_{\mathit{eff}} = \frac{M}{1 + (M-1)\bar{\rho}}} \tag{41}$$

$\bar{\rho}$*ICC* $\bar{\rho} \to 1 M_{\mathit{eff}} \to 1$——

Proof. $M\{v_1, \ldots, v_M\}\mathrm{Var}(v_m) = \sigma^2$ $\mathrm{Cov}(v_m, v_{m'}) = \rho\sigma^2 m \neq m'$ $\bar{v} = \frac{1}{M}\sum v_m$

$$\mathrm{Var}(\bar{v}) = \frac{\sigma^2}{M} + \frac{M-1}{M}\rho\sigma^2 = \frac{\sigma^2}{M}(1 + (M-1)\rho) \tag{42}$$

$\rho = 0 M/(1 + (M-1)\bar{\rho})$ Hoeffding □ □

[] $M_{\mathbf{eff}}$
$M_{\mathrm{eff}} M_{\mathrm{nominal}}$10%–30% “$M = 100$”10–30

6.2 ——Gödel-Turing

Remark 6.2 (Gödelian)*.* SCX**??** SCX$\Delta_s^{\mathrm{claimed}}\mathcal{V}$SCX-

(1) SCX“SCX”SCX

(2) SCX

(3) SCX——SCXSCX

Gödel

[] SCX SCXSCX**??**–**??** SCX—— Gödel-Turing

7 ——SCX

7.1

(1) **§2** “”——

(2) **§3**—— Fano

(3) **§4**—— “”

(4) **§5**——

SCXΔ_s

Table 1: SCX

		SCX
Popper		$\Delta_s^{\text{claimed}}\mathcal{V}$??
Kuhn		= Fano ??
Lakatos	++	$\mathcal{S}_K$ + ??3
Condorcet		??
Feyerabend	“”	$M_{\text{eff}} \to 1$—— “”??
IOANNIDIS (2005)	“”	??

7.2

7.3

??

—— $\bar{\rho}$ $p_{\text{clean},s}$

?? (i)

(ii) $\exp(-2M\Delta_s^2)$——Δ_s

7.4

—— ——SCX

[][] []

(1) **Gödel??**

(2) **??**

(3) **??** $H(\mathcal{S}_{\text{new}} \mid \mathcal{S}_{\text{old}})$——

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SCX(SCX Project, 2026) “” “”SCX SCX—— “”

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